

# Veer Phatak

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## PROFILE

Final-year MEng Aeronautics and Astronautics student specialising in Aerodynamics at the University of Southampton, graduating July 2027 with a 71.7% first-class Year 3 average. Team leader and aerodynamic design lead on a final-year project investigating the cornering aerodynamics of a MotoGP-style motorcycle through CFD and wind tunnel testing, also leading CAD development and model manufacture. Experienced across the aero development loop: ANSYS Fluent CFD, wind tunnel testing correlated against CFD and XFLR5, a first-class experimental fluid dynamics project with a bespoke computer-vision measurement system, and an R&D internship at TREND Networks where a redesigned part was approved for manufacture.

## CORE SKILLS

**Aerodynamics and CFD:** ANSYS Fluent, XFLR5, XFOIL, OpenVSP/VSPAERO, panel methods (Python), aerofoil and boundary layer analysis, aerodynamic trade studies

**Wind tunnel and experimental:** wind tunnel testing, CFD and experiment correlation, test rig design and build, calibration, computer-vision measurement, uncertainty analysis

**CAD and manufacture:** SolidWorks, design for manufacture (injection moulding, additive manufacturing), rapid 3D-printed prototyping (Bambu Studio), fit and tolerance checks

**FEA:** ANSYS Mechanical, mesh convergence studies, stress and displacement interpretation

**Programming and data:** Python (NumPy, pandas, OpenCV, Matplotlib), MATLAB, Excel; data processing, visualisation and automation

## AERODYNAMICS PROJECTS

**Final-Year Group Design Project: Cornering Aerodynamics of a MotoGP-Style Motorcycle** Sep 2026 - Present  
*Supervisors: Dr David Marshall and Dr Renan Soares*

- **Team leader** of a student group investigating the cornering aerodynamics of a MotoGP-style motorcycle through CFD and wind tunnel testing, coordinating the team and liaising with supervisors.
- **Aerodynamic design lead**, responsible for aerodynamic design direction across the CFD and wind tunnel programme.
- **CAD development and manufacturing lead**, responsible for the CAD model and manufacture of the wind tunnel test model.

**Individual Master's Project: Hydrodynamic Analysis of Large Plastics in Open-Channel Flow** Oct 2025 - May 2026

*Graded 75% (First). Supervisor: Dr Sergio Maldonado*

- Designed and built a laboratory open-channel flume and paddlewheel drive (SolidWorks, Arduino stepper control) to reproduce turbulent flow, covering component selection, bearing and shaft mounting, sealing and camera illumination.
- Developed a Python/OpenCV measurement system (segmentation, contour fitting, nearest-neighbour tracking, Kalman filtering, homography calibration) reconstructing trajectories from 240 fps footage at **98.4% mean detection coverage and 1.5 mm positional accuracy**.
- Analysed **58 experimental runs** across spheres, rods and discs using velocity, slip, drag, dispersion and tortuosity, with Reynolds and Stokes number analysis comparing flume and water tunnel results.
- Diagnosed sealing, drive response and frame-rate issues, and validated every tracked run against raw footage to reject reflections and missed detections.

**Fixed-Wing UAV Wing Design, Aerodynamics and Stability Lead (Group Project)** 2024 - 2025

- **Ranked 1st in the cohort:** delivered the widest minimum-to-maximum speed range in the year (23.1 m/s), over 10% more than any other team, using flaps and ailerons.
- Compared planforms in XFLR5 and XFOIL within mass and volume limits, balancing lift, drag and stability under load.
- Led configuration selection using decision matrices, and managed Gantt and resource planning that delivered the project early.

## EDUCATION

## University of Southampton, MEng Aeronautics and Astronautics (Aerodynamics)

Sep 2023 - Jul 2027

- **Year 3: 71.7% average (First).** Aerospace Control Design 80%, Individual Project 75%, Aircraft Structural Design 75%, Advanced Aeronautics 74%, Aerothermodynamics 68%.
- **Wind tunnel laboratories:** measured NACA 0020 pressure distribution against angle of attack, boundary layer development and finite wing lift and drag, and correlated results against CFD, XFOIL and XFLR5, analysing discrepancies.
- **Advanced Aeronautics:** implemented a Python doublet panel method with accuracy and panel-distribution convergence studies.
- **FEA:** ANSYS Mechanical mesh convergence study from roughly 24,000 to 1.9 million elements, balancing peak stress accuracy against solve time and memory.

## Reading School, Reading

Sep 2016 - Jul 2023

- A-levels: A\* Economics, A Mathematics, A Physics, B Classical Civilisation. AS Additional Mathematics (A). 10 GCSEs including eight grade 9s.

## ENGINEERING EXPERIENCE

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### TREND Networks, R&D Intern

Jul 2026 - Sep 2026

- **Handheld instrument enclosure:** designed from initial concept around a pre-manufactured PCB, fitting clearances, port positions and mounting points while reconciling requirements from several engineers. Refined over three iterations from a large, angular first version to a smaller, smoother design with protection for the test connections.
- **Battery compartment redesign:** redesigned an existing product's battery compartment for user-replaceable batteries under EU regulations within a fixed envelope, balancing battery protection, durability, styling and ease of use. Ten iterations produced a design **approved and released for manufacture**.
- **Ribbon fibre spool:** designed a spool to feed and protect fibre during testing, maximising stored fibre length within each tester case over three iterations.
- Ran all office 3D printing, prototyping each iteration the same day to check fit, wall thickness, warping and tolerances against the CAD, with injection moulding in mind for production parts.

### Southampton University Formula Student (SUFST), Race Engineering and Chassis

2023 - Present

- Researched front roll hoop integration into the chassis and presented technical and feasibility findings to the department and team management.
- Built Python and MATLAB lap-time and tyre degradation simulations to evaluate setup trade-offs, contributing to **2nd in the FSUK Simulator Championship and 10th overall at FSUK 2025**.

## LEADERSHIP AND WORK EXPERIENCE

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### Head Coach, Boyne Hill Cricket Club

May 2025 - Present

- Plan and lead weekly two-hour training sessions for U9 girls and U13 boys teams.

**Other experience:** Private Tutor, ages 8 to 14 (2022 - 2025); Front of House, Seasonality Restaurant (2022 - 2024); Dynamos Cricket Coach, ECB programme (2021 - 2022); Tutor, Brilliant Tutors Academy (2022); Sales Assistant and Till Manager, Kay's Chemist (2021 - 2022).

## ACHIEVEMENTS AND INTERESTS

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- 2nd overall, FSUK lap time challenge and Simulator Championship 2025.
- Boyne Hill CC 1st XI wicketkeeper-batter with 16 years of competitive cricket; captained Southampton University CC indoor team and outdoor 3rd XI.
- ECB and Berkshire qualified cricket coach; qualified first aider; Sports Prefect at Reading School.

*References available on request.*